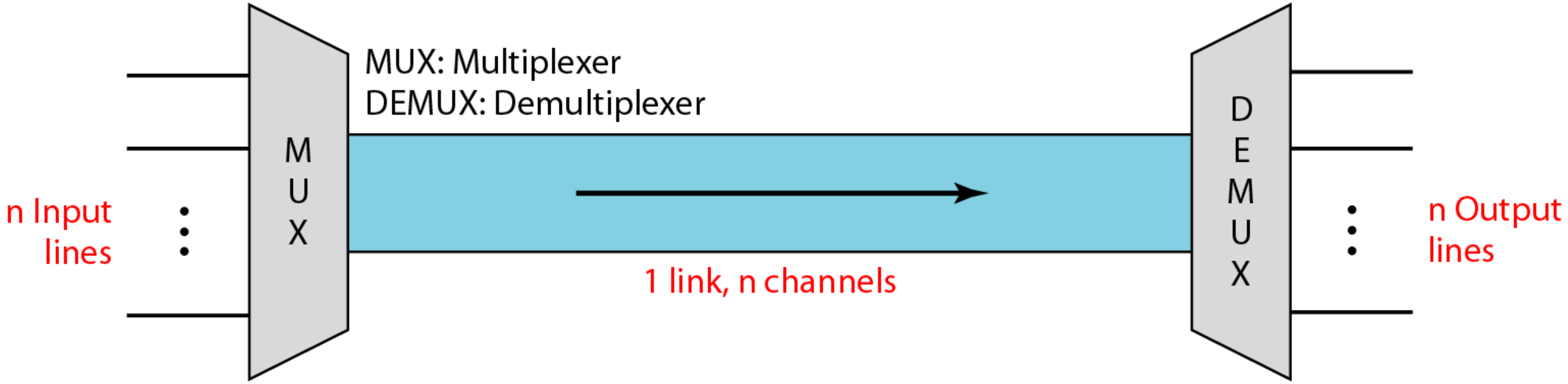
CHAPTER 8

Multiplexing

"It was impossible to get a conversation going, everybody was talking too much."

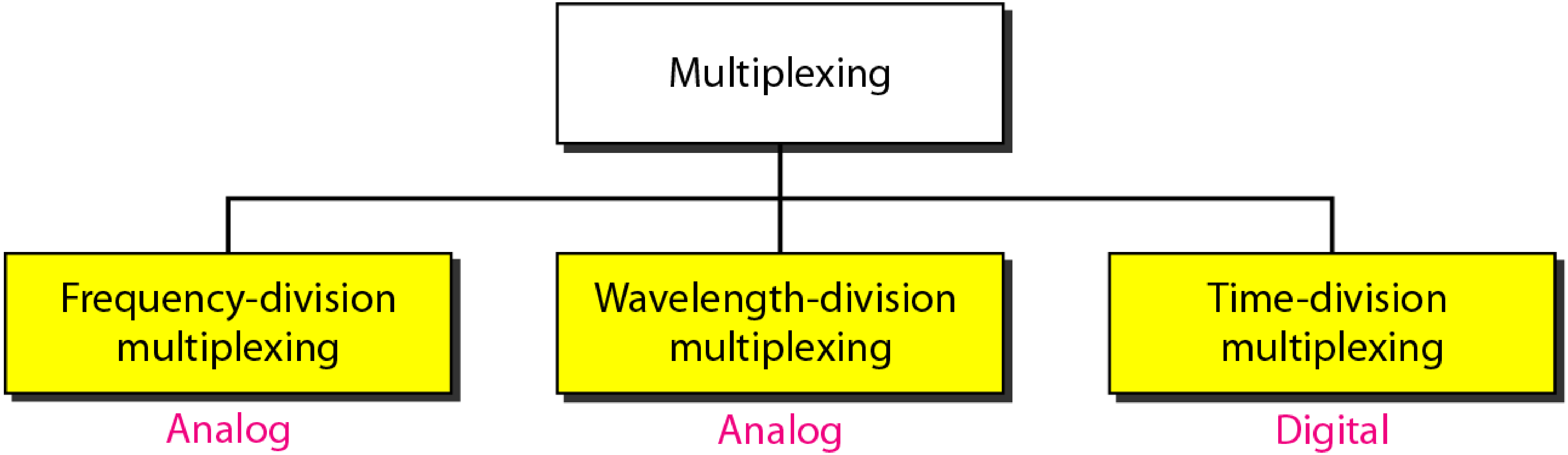
- Yogi Berra

MULTIPLEXING



Bandwidth utilization is the wise use of available bandwidth. Efficiency can be achieved by multiplexing; i.e., sharing of the bandwidth between multiple users.

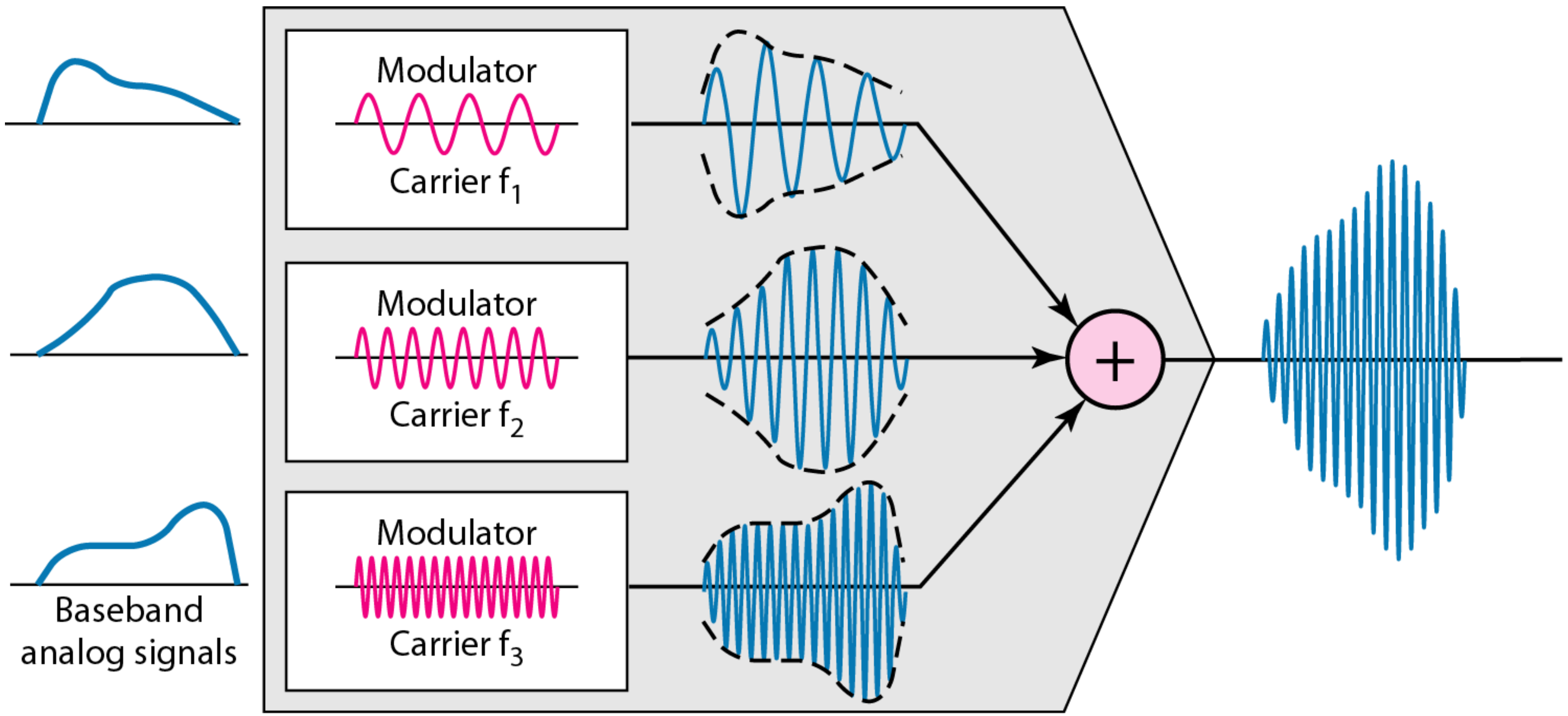
Categories of Multiplexing



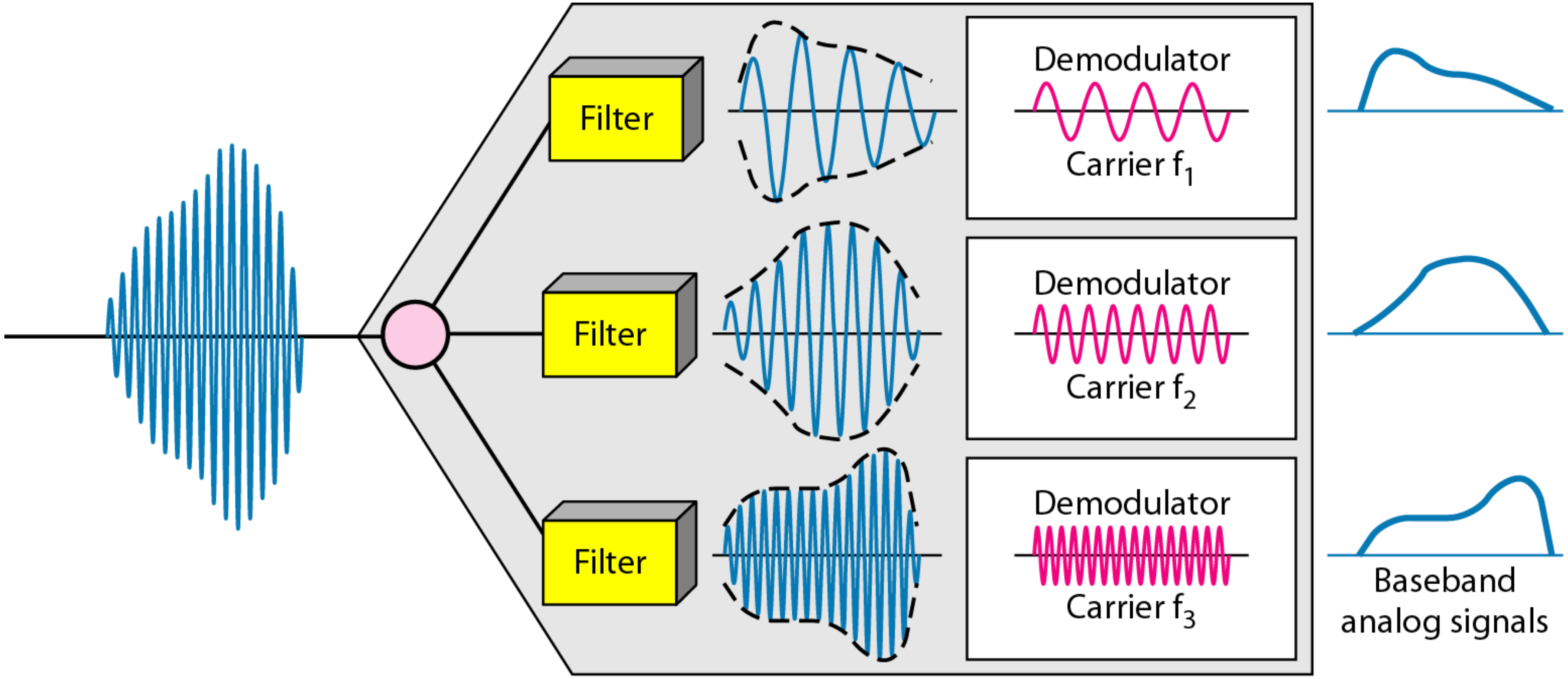
Frequency-division multiplexing (FDM)

FDM is an analog multiplexing technique that combines analog signals.

1- Frequency-division multiplexing (FDM) Process



FDM demultiplexing example



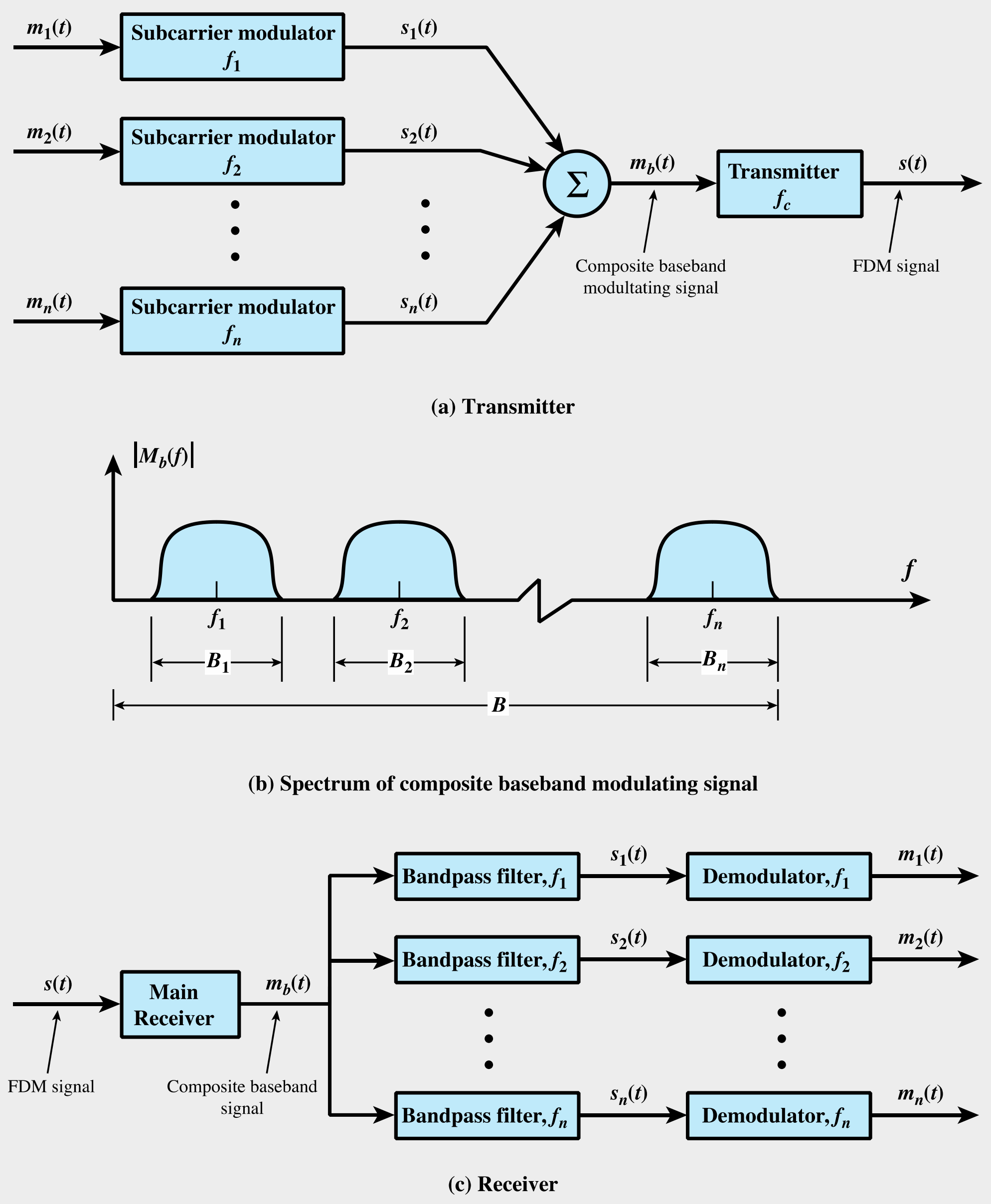


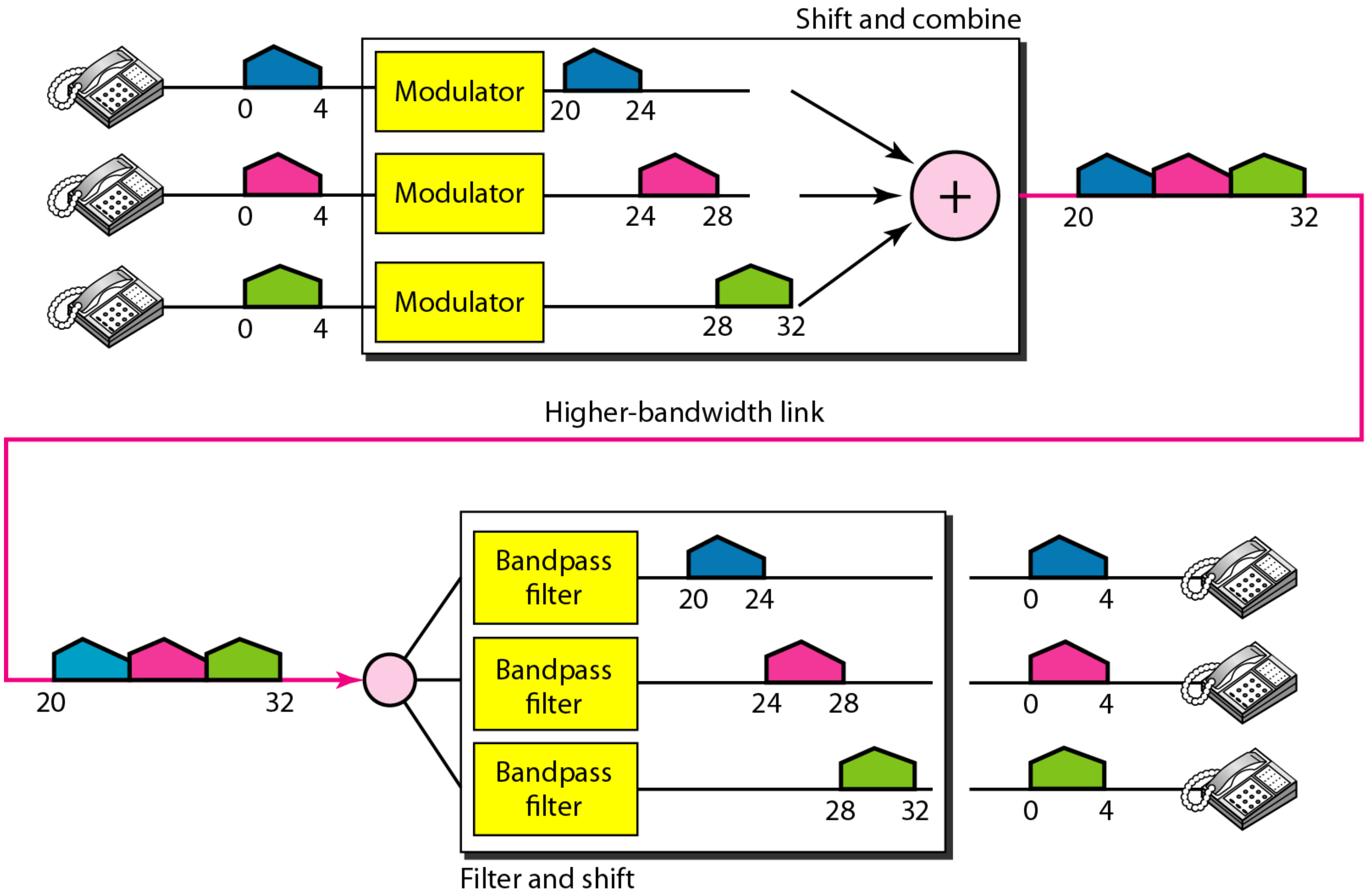
Figure 8.3 FDM System

Example

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.

Solution

We shift (modulate) each of the three voice channels to a different bandwidth. We use the 20- to 24-kHz bandwidth for the first channel, the 24- to 28-kHz bandwidth for the second channel, and the 28- to 32-kHz bandwidth for the third one.



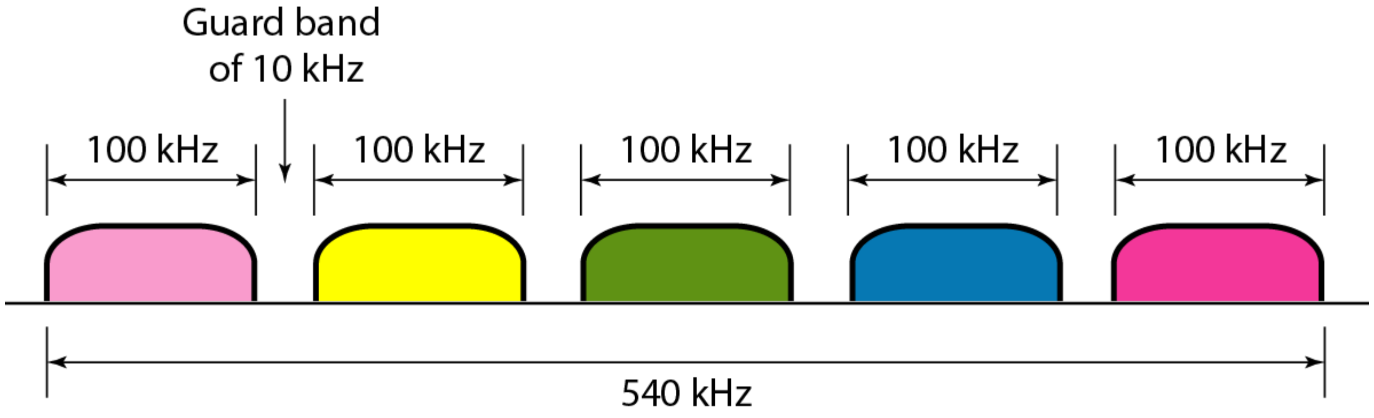
Shift and combine
Modulator
Modulator
Modulator
0
4
0
4
0
4
20
24
24
28
28
32
20
32
Higher-bandwidth link
20
32
Bandpass filter
Bandpass filter
Bandpass filter
20
24
24
28
28
32
0
4
0
4
0
4
Filter and shift

Example

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

Solution

$$5 \times 100 + 4 \times 10 = 540 \text{ kHz},$$



2- Wavelength Division Multiplexing (WDM)

- WDM is an analog multiplexing technique.
- Multiple beams of light at different frequencies
- Carried over optical fiber links
- Dense Wavelength Division Multiplexing (DWDM)

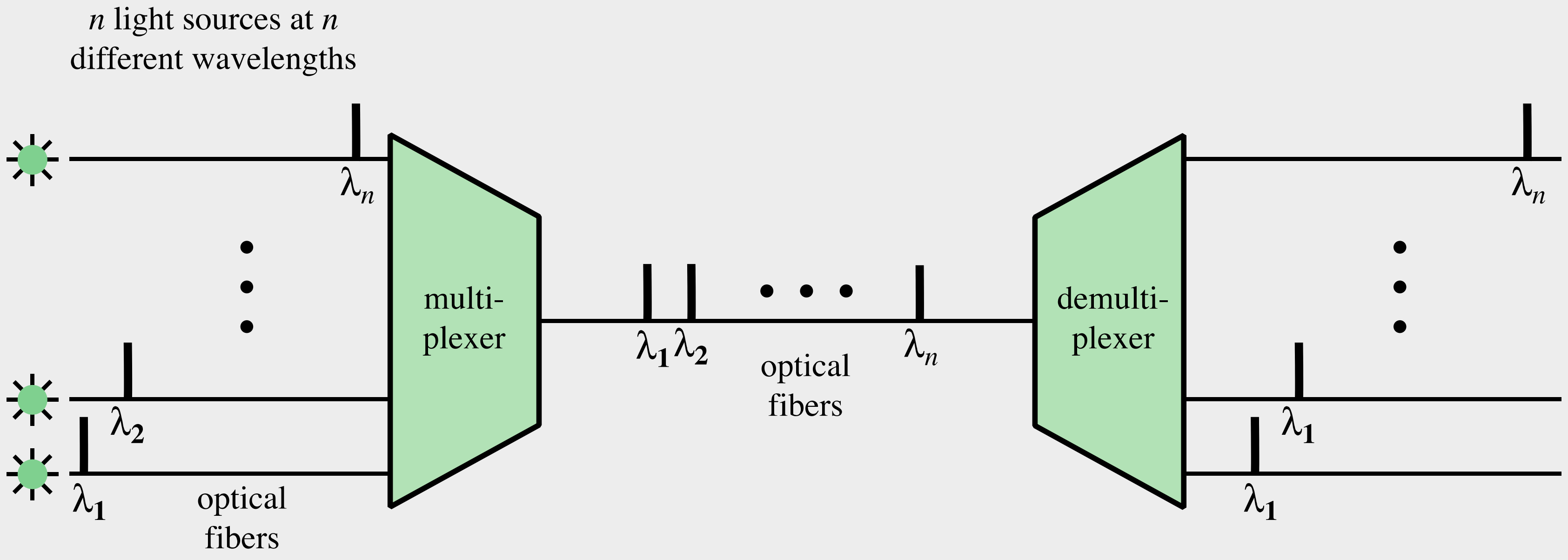
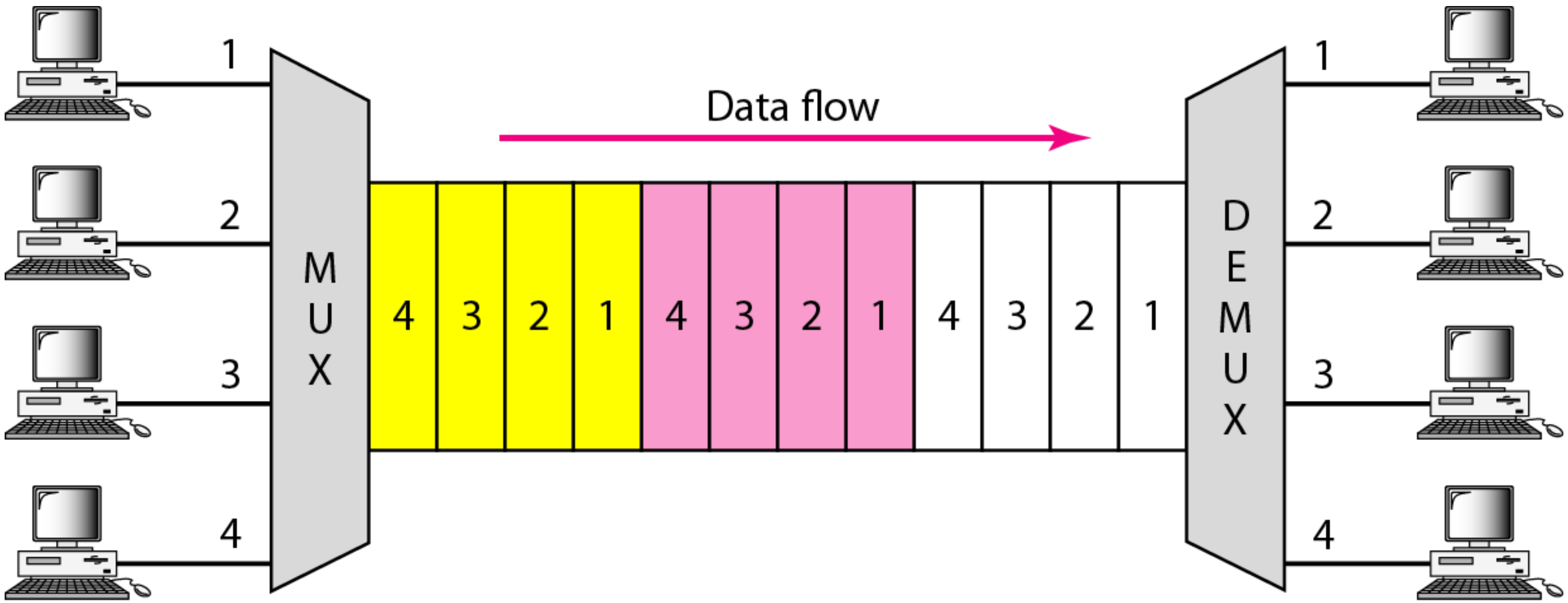


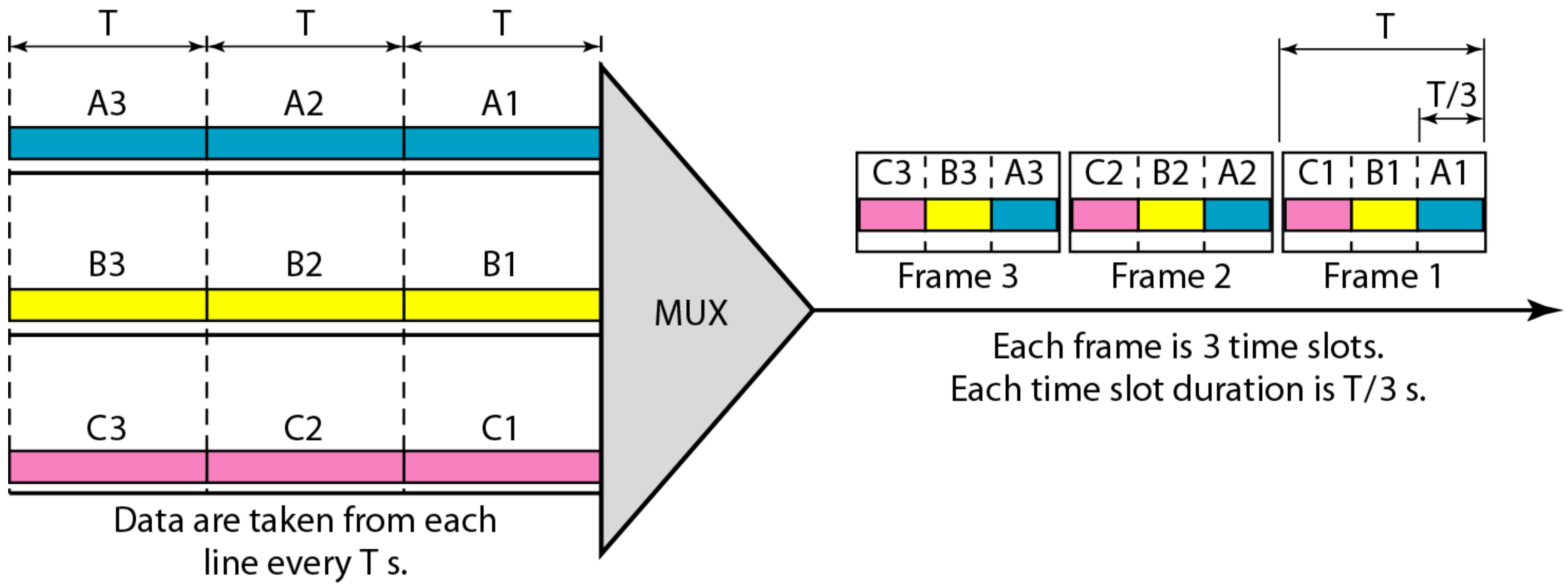
Figure 8.5 Wavelength Division Multiplexing

3- Time Division Multiplexing (TDM)



TDM is a digital multiplexing technique for combining several digital channels into one high-rate one.

Synchronous time-division multiplexing



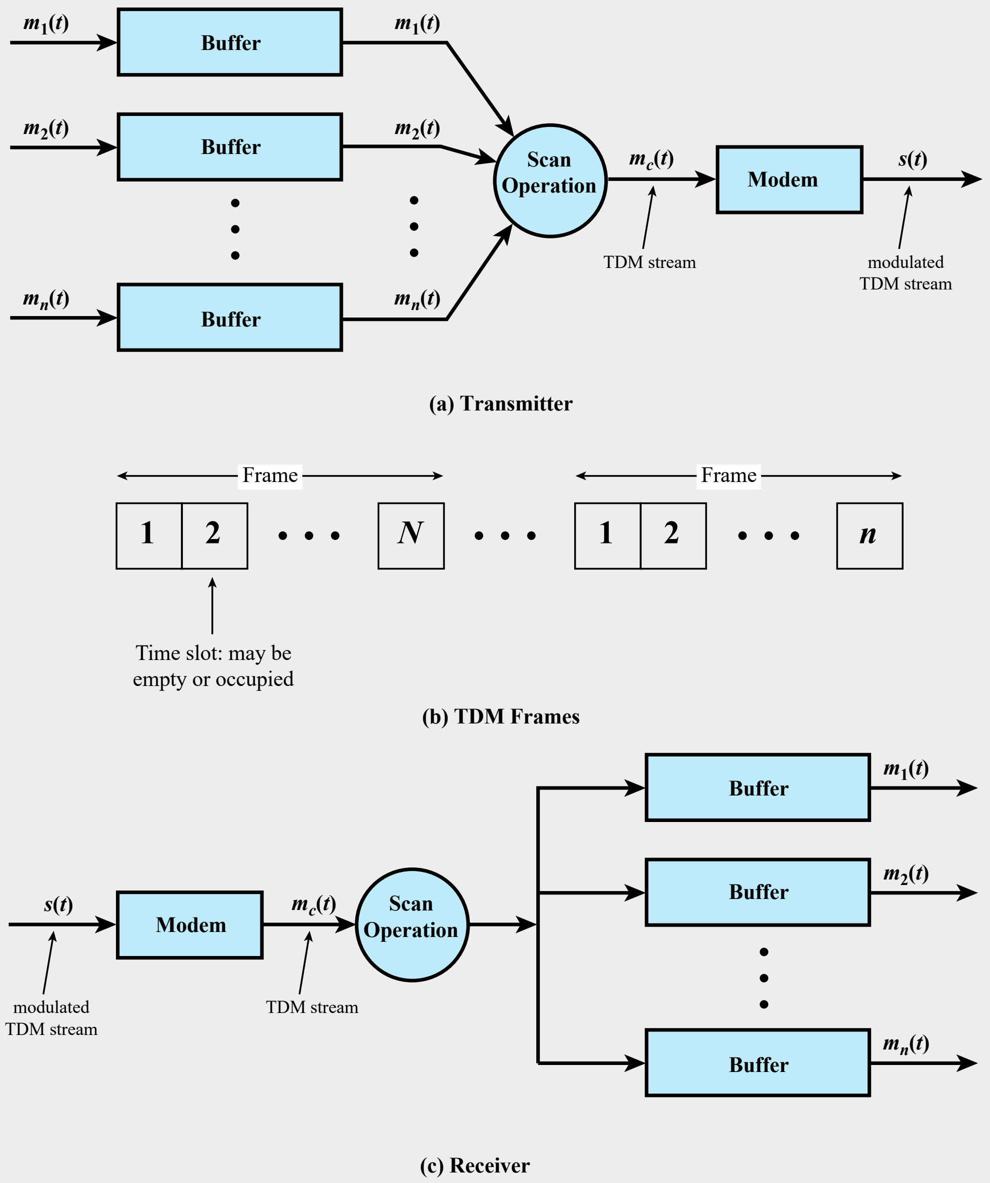
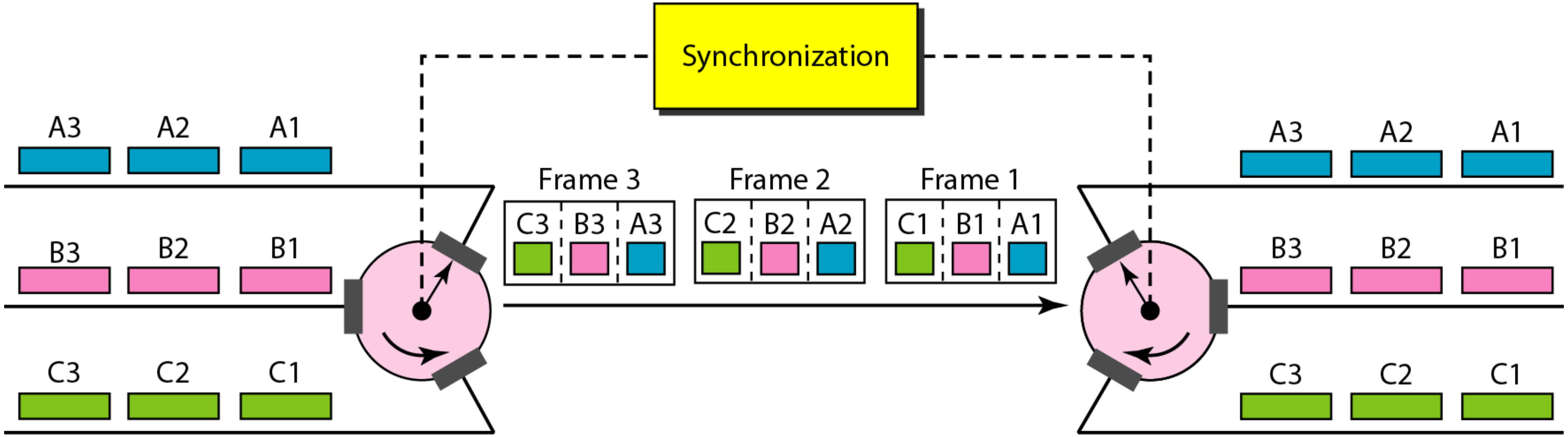


Figure 8.6 Synchronous TDM System

Interleaving

- The process of taking a group of bits from each input line for multiplexing is called interleaving.



Example

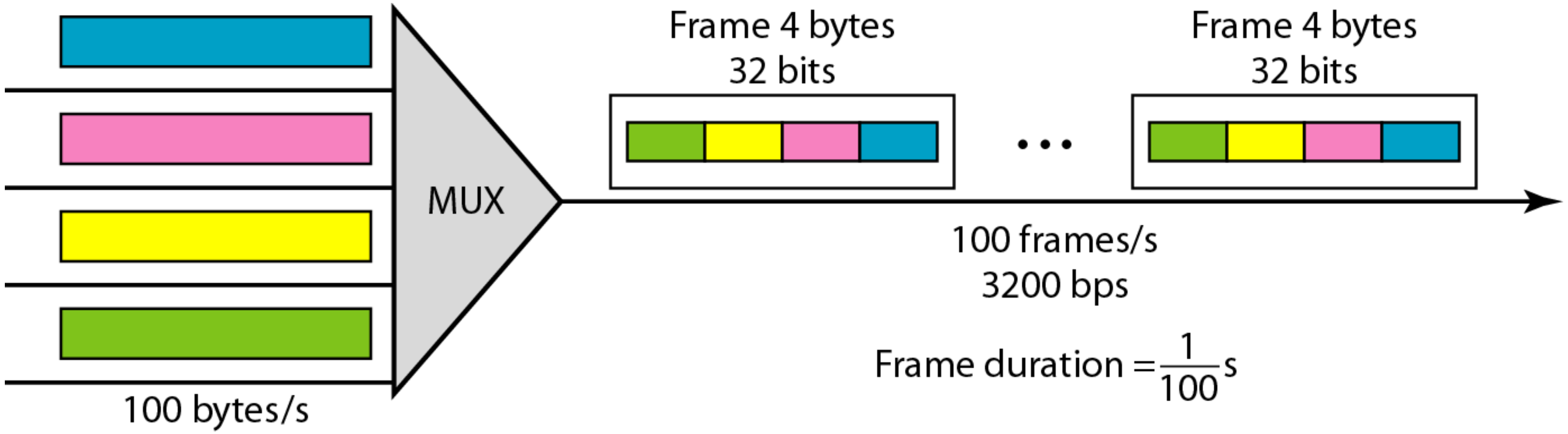
Four channels are multiplexed using TDM. If each channel sends 100 bytes /s and we multiplex 1 byte per channel, show

- *The frame traveling on the link,*
- *The size of the frame*
- *The duration of a frame, the frame rate, and*
- *The bit rate for the link.*

Solution

Each frame carries 1 byte from each channel;

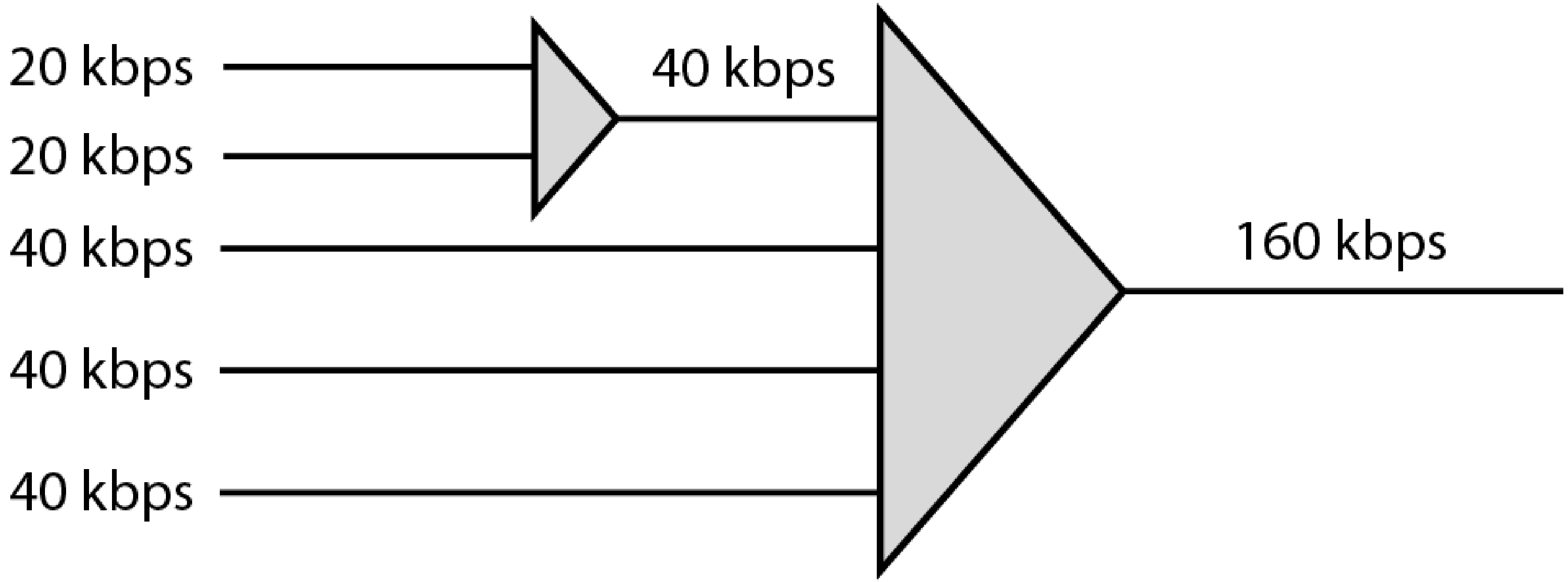
- *The size of each frame, therefore, is 4 bytes, or 32 bits.*
- *Because each channel is sending 100 bytes/s and a frame carries 1 byte from each channel, the frame rate must be 100 frames per second.*
- *Frame duration=1/ frame rate*
- *The bit rate is 100 × 32, or 3200 bps.*



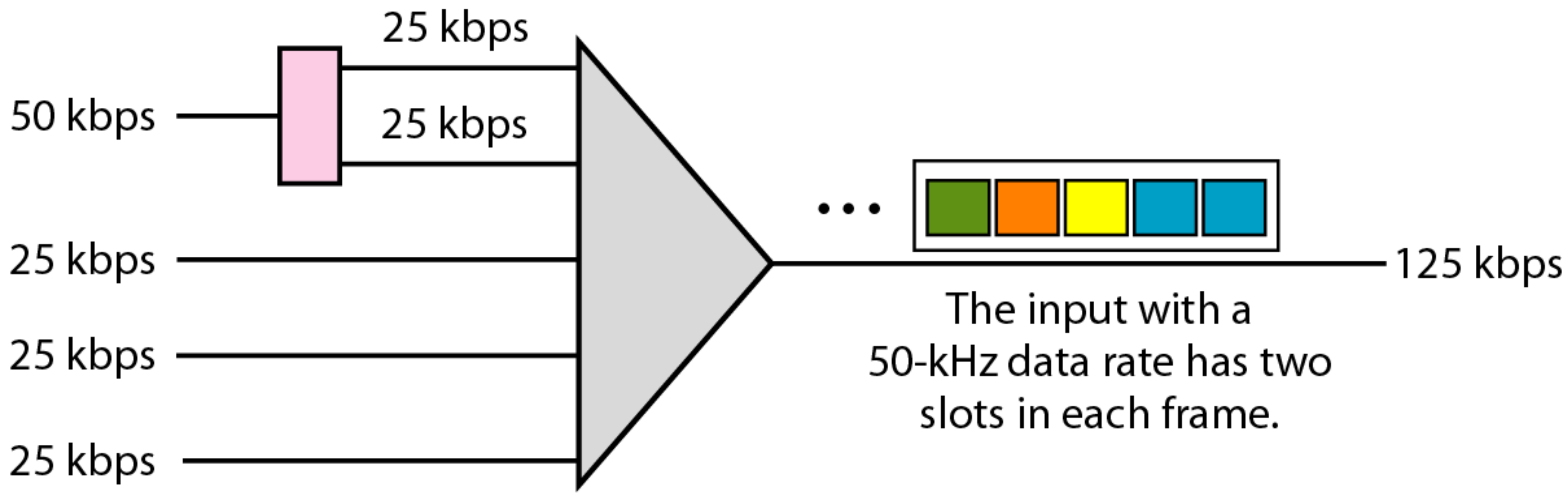
Data Rate Management

- Not all input links maybe have the same data rate.
- Some links maybe slower. There maybe several different input link speeds
- There are three strategies that can be used to overcome the data rate mismatch: multilevel, multislot and pulse stuffing

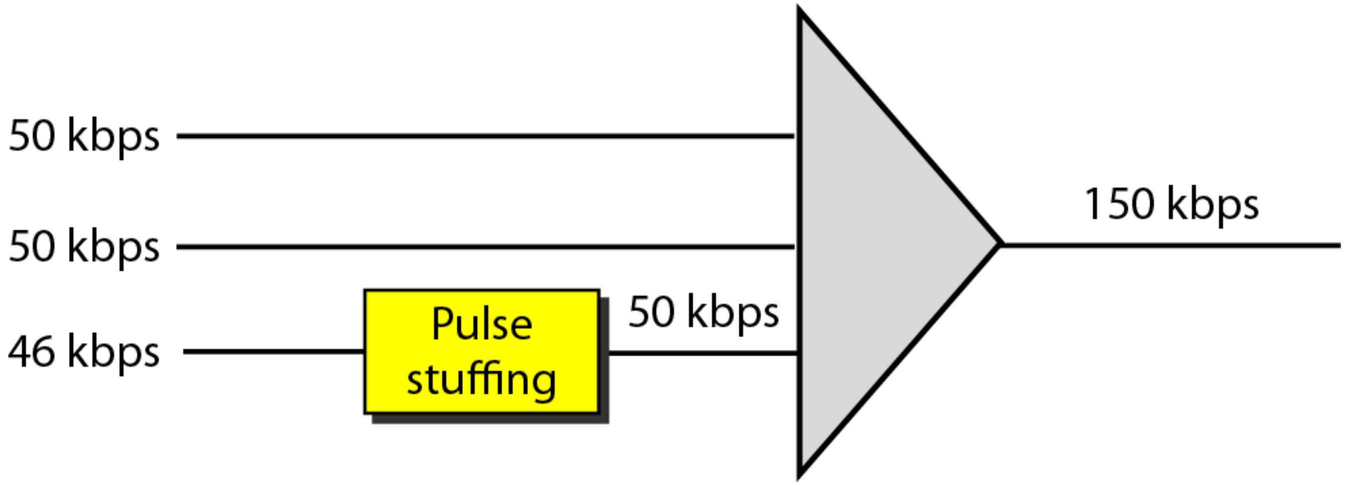
Multilevel multiplexing



Multiple-slot multiplexing



Pulse stuffing



Synchronization

- To ensure that the receiver correctly reads the incoming bits, i.e., knows the incoming bit boundaries to interpret a “1” and a “0”, a known bit pattern is used between the frames.
- The receiver looks for the anticipated bit and starts counting bits till the end of the frame.
- Then it starts over again with the reception of another known bit.
- These bits (or bit patterns) are called synchronization bit(s).
- They are part of the overhead of transmission.

Framing bits